

Example 1. Consider the first-order system

$$\dot{x} = -0.5x + u, \quad y = 2x.$$

Task 1. Write A , B , C , and D matrices for the system.

Task 2. Note the open-loop pole(s) of the system.

Task 3. Physically, we want the closed-loop output $y(t)$ to reach 63% of its final value in 1 second when a step input is applied. What is your desired pole?

Task 4. Find the state-feedback controller K and the feedforward gain k_f

$$u = -Kx + k_f r$$

that achieves your desired step response.

Example 2. Consider the DC motor transfer function

$$G(s) = \frac{2}{2s + 1}.$$

Task 1. Write a state-space representation in reachable canonical form.

Task 2. Note down A , B , C , and D .

Task 3. Note the open-loop pole(s) of the system.

Task 4. Physically, we want the closed-loop output $y(t)$ to reach 63% of its final value in 1 second when a step input is applied. What should the desired pole be?

Hint: For a first-order closed-loop system

$$G_{cl}(s) = \frac{b}{s + a},$$

a unit step input gives

$$Y(s) = \frac{b}{s(s + a)},$$

so

$$y(t) = y(\infty) (1 - e^{-at}).$$

Task 5. Find the state-feedback controller

$$u = -Kx$$

that achieves your desired pole.

Example 3. Consider the second-order system in reachable canonical form

$$\dot{x} = Ax + Bu, \quad y = Cx + Du$$

with

$$A = \begin{bmatrix} -2 & -13 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = [0 \quad 1], \quad D = 0.$$

Task 1. Run the open-loop step response in MATLAB:

$$\text{step}(\text{ss}(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}))$$

What do you observe about the oscillation frequency of the system?

Task 2. Note the open-loop poles of the system.

MATLAB hint:

$$\text{eig}(\mathbf{A})$$

Task 3. We want the closed-loop system to have a lower oscillation frequency than the open-loop system. Propose a pair of desired poles.

Task 4. Use MATLAB to compute the state-feedback gain:

$$\mathbf{K} = \text{place}(\mathbf{A}, \mathbf{B}, \mathbf{DP})$$

where $\mathbf{DP}$ is your vector of desired poles.

Task 5. With state feedback $u = -Kx$, write the closed-loop matrix

$$A_{cl} = A - BK$$

in terms of

$$K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}.$$

Task 6. Compare the entries of A and A_{cl} . Which coefficients change under state feedback? How does this connect to what we saw in controller canonical form in class?